

RECIP-e FHIR Cookbook

Version 1.0

Recip-e FHIR cookbook

Contents

DOCUMENT MANAGEMENT	3
1.1. History	3
1.2. Purpose	4
1.3. Support	4
2. ARCHITECTURAL PRINCIPLES	5
2.1. Pseudonymisation.....	5
2.2. Authentication.....	5
2.3. Authorization.....	8
3. FHIR INTERACTIONS.....	13
3.1. CapabilityStatement	13
3.2. OpenAPI Definition	13
3.3. Supported profiles.....	13
3.4. Use of pseudonyms	13
3.5. Error codes.....	14
4. HEADERS SUMMARY	17
4.1. Request Headers	17
4.2. Response Headers	17
5. BUSINESS INFORMATION	18
5.1. Business rules	18
5.2. Status flow.....	18
5.3. Privacy flags.....	19

Document management

1.1. History

Version	Status	Date	Author(s)	Changes
0.1	Draft	10.10.2025	Dries Vanstappen	First version
0.2	Draft	01.04.2026	Dries Vanstappen	First amendaments
0.3	Draft	20.04.2026	Dries Vanstappen	Addition of business rules and status flow
1.0	Official version	04.06.2026	Dries Vanstappen, Hendrik Janssen	FHIR-A-THON feedback

1.2. Purpose

This cookbook is for developers, analysts and software producers involved in the integration into their software solution of the services delivered by Recip-e, specifically for the FHIR data.

Fast Healthcare Interoperability Resources (FHIR) is a next generation standards framework created by HL7. Solutions are built from a set of modular components called “Resources”. These resources can easily be assembled into working systems that can solve clinical and administrative problems. More about FHIR can be read on the FHIR website: <http://hl7.org/fhir/>.

FHIR has defined general, international definitions of resources. Nevertheless, within the standard, it is possible to create a local variant based on those international resources. In a co-creation with the industry, the healthcare-sector, HL7 and the different governments involved eHealth has created a Belgian version of international resources. These Implementation Guides can be retrieved on the eHealth website: <https://www.ehealth.fgov.be/standards/fhir/index.html>.

The goal of the cookbook is to document how a software can accomplish a technical integration with Recip-e in FHIR.

The business use cases and data models themselves are not part of this cookbook.

This document is composed of three sections:

- Section [Principles](#) explains the architectural principles as authentication, authorization and security.
- Section [FHIR interactions](#) provides an overview of the supported interactions on the server.
- Section Examples of CRUD operations presents the structure of the requests to perform CRUD operations on the server.

The latest version of this CookBook can always be found at <https://github.com/smals-belgium/shared-digital-medication-prescription>.

1.3. Support

For issues in both acceptance and production, please use the form on technicalsupport.recip-e.be to contact us. In order to guarantee the best possible support, please provide as many details when reporting an issue: environment, timestamp, URL, method, request parameters/body, response code/body,

2. Architectural principles

Recip-e has selected some specifications in order to guarantee the regulations on the data shared in the ecosystem.

2.1. Pseudonymisation

To ensure privacy regulation, Recip-e has chosen to use pseudonyms for identifiers of patients in order to guarantee security around the medical data of each patient that is stored in the database.

To accomplish pseudonymization, eHealth's pseudo service, as trusted party, will be used to pseudonymize, convert and identify data.

Recip-e will expect incoming data to contain pseudonyms of specific data elements. Please refer to [Use of pseudonyms](#) on how to accomplish this in FHIR. Software integrators will need to onboard with eHealth to be able to pseudonymize data using the pseudo service. Access needs to be granted for the software package so it can pseudonymize, identify and/or convert to the UHMEP domain.

More details can be found on the eHealth documentation:

- [Pseudonymisering & Anonimisering \(NL\)](#)
- [Pseudonymisation & Anonymisation \(FR\)](#)

Technical information about the UHMEP domain can be retrieved here (a valid token is required):

- [uhmep_v1_domain \(ACC\)](#)
- [uhmep_v1_domain \(PRD\)](#)

Smals recommends usage of the Pseudonymisation Helper library:

- [Java Library](#)
- [.NET Library](#)
- [Javascript Library](#)

2.2. Authentication

2.2.1. eHealth IAM

Recip-e expects tokens from the authentication service of eHealth: I.AM Connect.

The Identity Access Management (IAM) Connect service is an authentication service provided by the eHealth platform. Its goal is to gather the necessary information to authenticate and authorize traffic coming from a server, granting it in a secure way the required access to utilize certain services, such as the Recip-e project.

eHealth IAM uses the [OpenID Connect](#) layer on top of the [OAuth2.0 protocol](#), where it provides a JWT token for the client who can use it to identify itself at various eHealth services. The granted access token should be put in the Authorization header of each REST request to Recip-e as a Bearer token.

Prescribers and citizens should use the healthcare realm to obtain their tokens. Pharmacists should use the M2M (Machine-to-machine) realm with the pharmacy client. In hospital context, prescribers use the M2M realm with the hospital client. Note: hospital pharmacies use their own M2M client, they cannot reuse the hospital's client.

More information about I.AM Connect can be found here:

- [I.AM \(Identity & Access Management\) \(NL\)](#)
- [I.AM \(Identity & Access Management\) \(FR\)](#)

2.2.2. eHealth IAM Client Onboarding

A client should register itself in the healthcare or M2M realm. The following client registration document should be filled out and sent to eHealth to get access:

- [I.AM Connect - HealthCare Client Registration Form](#)
- [I.AM Connect - M2M Client Registration Form](#)

Each healthcare professional or organization must register upfront and gain access to the platform for managing and accessing patient prescriptions, requiring human authentication and identity verification.

Please find the list of needed information that would need to be provided below (current version 2.1).

2.2.2.1. General Client Information

Information (fields with an * are mandatory)	Explanation and allowed values	Example
Request date (*)	Please state the date on which you are submitting this request form.	01/03/2025
Partner organization (*)	Please state the full name of the partner organization that is requesting the IAM onboarding.	Aqme Care
Contact person (*)	Please state the full name, email address and phone number of the person that may be contacted by eHealth for information on the onboarding request. Note that this contact person may be contacted for questions related to business as well as technical aspects, so the contact person is expected to coordinate with all relevant departments within your organization.	Name: John Doe Email address: john.doe@aqmecare.be Feel free to add additional email addresses or a group mail address in case more people need to be kept informed. Phone number: 0470/12.34.56
Brief description of the purpose of your application and of the requested client. (*)	Please describe briefly what the purpose of your application is.	Aqme care is currently building an application so patient can access to their prescriptions on the way.....
Public or confidential client? (*)	Please indicate which of the two available authentication flows your application uses (select only one option): Public client: Your application is either a distributed software, or a webapp that runs 100% on the client of the end user. (It is not possible in this case to generate a private key to authenticate the application.) The user authenticates directly with eHealth IDP and FAS. Authorization code flow: PKCE integration. Confidential client: Your application runs partly on the	Public client

	server(s) of a recognized partner organization (private keys are used to authenticate the application). The user authenticates directly with eHealth IDP and FAS. Authorization code flow: an access token is sent by the client-component to the server component of the recognized organization.	
Client ID (*)	The IAM client ID is the unique identifier of the IAM Connect client configured by eHealth for the partner. In case the partner organization already has an existing IAM Connect Healthcare client, the partner can choose (or not) to request for the existing client to be reused and adapted. (In some cases, though this is not possible, and in that case, eHealth will have to configure a new client.)	software-name
Scopes	Scopes are boundaries that are defined to technically limit the use of the IAM client to the purpose/application for which it is requested. In case your request for an IAM client is in the context of an onboarding procedure for a specific application, check the onboarding documentation of that application for guidelines on scopes to be filled out in this field. If the documentation mentions no scopes, leave the field empty.	• openid • iam:connect:exchange • iam:authz • ssin • profile • roles • pseudo:api:pseudonymize • pseudo:api:identify • nihdi:uhmep:pseudo • nihdi:uhmep:hcp
Redirect URI (*)	To redirect the user after a successful authentication, a valid redirect URI is needed in the configuration. This URI is also used for redirecting the user after a logout.	https://app-acc.software-name.be/iam-connect-endpoint/

2.2.2.2. Additional Confidential Client Information

This is only to be filled out if you have selected the option "Confidential client" higher in this form. This information does not apply to public clients. It is related to the credentials of the eHealth certificate used by your organization.

Information (fields with an * are mandatory)	Explanation and allowed values	Example
Type (*)	Please specify the type of identifier mentioned in your eHealth certificate.	NIHII-HOSPITAL (hospital) NIHII-PHARMACY (pharmacy) CBE (institution)
Identifier (*)	Please state the value of your certificate identifier (string value).	Type of identifier (e.g. "NIHII institution")
Application ID	If your eHealth certificate contains an application ID, please state it here. If not, leave this field empty	Application ID (if available)

2.3. Authorization

2.3.1. JWT Tokens

The JWT token that the client provides for professional and organizational users should be self-contained: the information provided in the JWT is enough to perform all authorization checks.

In case of citizen tokens, the UserInfo endpoint will always be called to retrieve pseudonymized personally identifiable information.

Recip-e supports only eHealth tokens coming from the IAM Connect healthcare or M2M realm.

The following OIDC configuration should be used by your application:

- issuer-uri (healthcare):
 - ACC: <https://api-acpt.ehealth.fgov.be/auth/realms/healthcare>
 - PROD: <https://api.ehealth.fgov.be/auth/realms/healthcare>
- issuer-uri (m2m):
 - ACC: <https://api-acpt.ehealth.fgov.be/auth/realms/m2m>
 - PROD: <https://api.ehealth.fgov.be/auth/realms/m2m>
- client-id: <your_client_id>
- scope: openid
- redirect-uri: <your_redirect_URI>
- supported modes: authorization code, refresh token

The retrieved access token should be sent to Recip-e with the header:

```
Authorization: Bearer <access_token>
```

2.3.2. Token Exchange

The use of the IAM Token exchange Flow is a security prerequisite for the use of the UHMEP FHIR API in the context of Healthcare realm client authentication. The initially requested IAM token will give users access to the UHMEP FHIR API, but also has the rights of these users to the pseudonymization service. More specifically, it will contain the right to identify and pseudonymize information. For security reasons, technical providers (such as Smals) cannot access these pseudonymization rights.

To avoid that from happening, the IAM Token exchange has been put in place. Software Integrators will exchange the token obtained initially at the user connection for another token to call the UHMEP FHIR API. In this token exchanged, the right for the pseudonymization has been removed.

Note: for healthcare providers, the plain text SSIN will still be part of the exchanged token, but for patients, the SSIN is stripped. Recip-e will retrieve a pseudonymized patient SSIN from the UserInfo endpoint, given the access token.

Technically, in order to perform the token exchange, a POST must be sent to this endpoint:

- ACC: <https://api-acpt.ehealth.fgov.be/auth/realms/healthcare/protocol/openid-connect/token>
- PROD: <https://api.ehealth.fgov.be/auth/realms/healthcare/protocol/openid-connect/token>

with the following form body:

```
requested_token_type=urn:ietf:params:oauth:token-type:access_token
grant_type=urn:ietf:params:oauth:grant-type:token-exchange
subject_token_type=urn:ietf:params:oauth:token-type:access_token
```

```
subject_token=<original_login_access_token>
client_id=<source_client_id>
audience=<target_audience>
```

Integrators of patient software should use “nihdi-uhmep-fhir-patient” as audience, integrators of software for professionals should use “nihdi-uhmep-fhir-hcp” as audience.

2.3.3. Token Claims

The app reads these claims from the authenticated professional user:

- ssin
- userProfile

The “userProfile” claim is expected to contain professional or organisation entries such as physician, nurse, dentist, midwife, pharmacist, pharmacy, or hospital, each with nihii11 (professional) or nihii8 (organizational) identifiers.

Note: currently, only the professional userProfiles are currently supported. This means that pharmacists can only connect when they’re personally authenticated, not via their pharmacy organization. This will be changed in a later version of the software to only support pharmacy organizations.

2.3.3.1. *Prescriber Token Example*

Initial Token

```
{
  "exp": 1778507034,
  "iat": 1778506734,
  "auth_time": 1778506359,
  "jti": "onrtac:2669acb5-1bd2-4374-8bbd-e0330992660d",
  "iss": "https://api-acpt.ehealth.fgov.be/auth/realms/healthcare",
  "aud": [
    "integrator-a-azp",
    "nihdi-uhmep-api",
    "ehealth-pseudo-api"
  ],
  "sub": "0837e8d5-7a4a-469c-bb4f-0682cdf5272a",
  "typ": "Bearer",
  "azp": "integrator-a-azp",
  "sid": "3f9e8234-9966-4b57-9a42-49434e1f2c84",
  "allowed-origins": [
    "https://app-acc.software-name.be/iam-connect-endpoint"
  ],
  "resource_access": {
    "nihdi-uhmep-api": {
      "roles": [
        "pseudo"
      ]
    },
    "ehealth-pseudo-api": {
      "roles": [
        "identify",
        "pseudonymize"
      ]
    }
  },
  "scope": "openid profile ssin iam:connect:exchange pseudo:api:identify pseudo:api:pseudonymize nihdi:uhmep:pseudo iam:authz",
  "ssin": "90010212345",
  "name": "John Doe",
  "preferred_username": "90010212345",
  "locale": "nl",
```

```

"given_name": "John",
"family_name": "Doe",
"userProfile": {
  "firstName": "John",
  "lastName": "Doe",
  "ssin": "90010212345",
  "physician": {
    "recognised": true,
    "nihii11": "19766016004"
  }
}
}

```

Exchanged Token

```

{
  "exp": 1778507087,
  "iat": 1778506787,
  "auth_time": 1778506359,
  "jti": "onrtac:2669acb5-1bd2-4374-8bbd-e0330992660d",
  "iss": "https://api-acpt.ehealth.fgov.be/auth/realms/healthcare",
  "aud": [
    "integrator-a-exchanged-azp",
    "nihdi-uhmep-api"
  ],
  "sub": "0837e8d5-7a4a-469c-bb4f-0682cdf5272a",
  "typ": "Bearer",
  "azp": "integrator-a-exchanged-azp",
  "sid": "3f9e8234-9966-4b57-9a42-49434e1f2c84",
  "allowed-origins": [
    "https://unavailable"
  ],
  "resource_access": {
    "nihdi-uhmep-api": {
      "roles": [
        "prescriber",
        "restricted"
      ]
    }
  },
  "scope": "openid profile nihdi:uhmep:hcp iam:authz",
  "act": {
    "azp": "integrator-a-azp"
  },
  "name": "John Doe",
  "preferred_username": "90010212345",
  "locale": "nl",
  "given_name": "John",
  "family_name": "Doe",
  "userProfile": {
    "firstName": "John",
    "lastName": "Doe",
    "ssin": "90010212345",
    "physician": {
      "recognised": true,
      "nihii11": "19766016004"
    }
  },
  "client_id": "integrator-a-client-id"
}

```

2.3.3.2. Pharmacist Token Example

Note: Pharmacists are temporarily allowed personally to Recip-e, until Pharmacies as an organization are supported.

The initial and exchanged tokens are similar to the prescriber example, except for the userProfile claim:

```
"userProfile": {
  "firstName": "John",
  "lastName": "Doe",
  "ssin": "90010212345",
  "pharmacist": {
    "recognised": true,
    "nihii11": "20017534001"
  }
}
```

The exchanged token also has this claim:

```
"resource_access": {
  "nihdi-uhmep-api": {
    "roles": [
      "caregiver",
      "restricted"
    ]
  }
}
```

2.3.3.3. Patient Token Example

Initial Token

The initial token is similar to the prescriber example, except for the claims:

```
{
  "aud": [
    "integrator-b-azp",
    "ehealth-pseudo-api"
  ],
  "azp": "integrator-b-azp",
  "scope": "openid pseudo:api:identify iam:connect:exchange pseudo:profile
pseudo:api:pseudonymize iam:profile:citizen iam:authz",
}
```

Exchanged Token

The exchanged token is similar to the prescriber example, except for the claims:

```
{
  "aud": [
    "integrator-b-exchanged-azp",
    "nihdi-uhmep-api"
  ],
  "azp": "integrator-b-exchanged-azp",
  "resource_access": {
```

```

    "nihdi-uhmep-api": {
      "roles": [
        "patient"
      ]
    },
    "scope": "openid iam:pseudo nihdi:uhmep:patient iam:authz",
    "act": {
      "azp": "integrator-b-azp"
    },
    "userProfile": {},
    "client_id": "integrator-b-client-id"
  }
}

```

Notes:

- Mandates are not yet supported in the current version of the application, but will come soon.
- For more information on the eHealth IAM.Connect claims, see the [claim mappers](#) document on the eHealth page. Recip-e only supports V1.

2.3.4. M2M Tokens

2.3.4.1. Requesting an M2M token

M2M clients are not able to do the token exchange as described above. They will either request the token with or without pseudonymisation scopes through:

- ACC: POST <https://api-acpt.ehealth.fgov.be/auth/realms/M2M/protocol/openid-connect/token>
- PROD: POST <https://api.ehealth.fgov.be/auth/realms/M2M/protocol/openid-connect/token>

With body :

```

grant_type=[client_credentials]
client_assertion_type=[urn:ietf:params:oauth:client-assertion-type:jwt-bearer]
client_assertion=<client_assertion>
client_id=[<client_id>]
scope=[openid iam:authz nihdi:uhmep:organization]

```

When requesting pseudonymization rights, the scopes change to:

```

scope=[openid iam:authz nihdi:uhmep:organization pseudo:api:pseudonymize
pseudo:api:identify nihdi:uhmep:pseudo]

```

The client assertion is a JWT that contains:

```

{
  "iss": "<client_id>",
  "sub": "<client_id>",
  "aud": [
    "https://api(-acpt).ehealth.fgov.be/auth/realms/M2M"
  ],
  "jti": "<random_uuid>",
  "iat": <issuer_at_time_epoch_seconds>,
  "exp": <expiration_time_epoch_seconds>
}

```

The client assertion must be signed with the private key associated with the M2M client's registered eHealth authentication certificate.

2.3.4.2. Executor Example

Example executor token with pseudonymization scopes:

```
{
  "exp": 1787927603,
  "iat": 1787927303,
  "jti": "trrtcc:8d3469bd-a1d0-79a8-b865-649621307b8b",
  "iss": "https://api-acpt.ehealth.fgov.be/auth/realms/M2M",
  "aud": [
    "nihdi-uhmep-api",
    "ehealth-pseudo-api"
  ],
  "sub": "3e81e464-841f-4b92-9bf5-6c7fa4d34624",
  "typ": "Bearer",
  "azp": "nihdi-pharmacy-20003873",
  "resource_access": {
    "nihdi-uhmep-api": {
      "roles": [
        "pseudo"
      ]
    },
    "ehealth-pseudo-api": {
      "roles": [
        "identify",
        "pseudonymize"
      ]
    }
  },
  "scope": "openid pseudo:api:identify iam:authz pseudo:api:pseudonymize
profile nihdi:uhmep:pseudo",
  "preferred_username": "nihdi-pharmacy-20003873",
  "userProfile": {
    "organizations": [
      {
        "pharmacy": {
          "nihii": "20003873",
          "name": "RECIP-E TEST PHARMACY"
        }
      }
    ]
  }
}
```

Example executor token without pseudonymization scopes:

```
{
  "exp": 1787927996,
  "iat": 1787927696,
  "jti": "trrtcc:f4ad5868-8da9-86ee-de6d-27d8a7e945a8",
  "iss": "https://api-acpt.ehealth.fgov.be/auth/realms/M2M",
  "sub": "3e81e464-841f-4b92-9bf5-6c7fa4d34624",
  "typ": "Bearer",
  "azp": "nihdi-pharmacy-20003873",
  "scope": "openid iam:authz profile",
  "preferred_username": "nihdi-pharmacy-20003873",
  "userProfile": {
    "organizations": [
      {
        "pharmacy": {
          "nihii": "20003873",
          "name": "RECIP-E TEST PHARMACY"
        }
      }
    ]
  }
}
```

```

    }
  ]
}

```

2.3.4.3. Hospital Example

Example hospital token with pseudonymization scopes:

```

{
  "exp": 1787928082,
  "iat": 1787927782,
  "jti": "trrtcc:ef188c73-abc0-4427-e76c-28e09b233dc1",
  "iss": "https://api-acpt.ehealth.fgov.be/auth/realms/M2M",
  "aud": [
    "nihdi-hospital-71089914-uhmep",
    "nihdi-uhmep-api",
    "ehealth-pseudo-api",
    "nihdi-pss-api"
  ],
  "sub": "36ee742e-68bd-4029-b3c6-a9323e43d63c",
  "typ": "Bearer",
  "azp": "nihdi-hospital-71089914-uhmep",
  "resource_access": {
    "nihdi-uhmep-api": {
      "roles": [
        "organization",
        "pseudo"
      ]
    },
    "ehealth-pseudo-api": {
      "roles": [
        "identify",
        "pseudonymize"
      ]
    },
    "nihdi-pss-api": {
      "roles": [
        "pss"
      ]
    }
  },
  "scope": "openid pseudo:api:identify iam:authz nihdi:pss
pseudo:api:pseudonymize nihdi:uhmep:organization profile nihdi:uhmep:pseudo",
  "clientHost": "ip_address",
  "preferred_username": "nihdi-hospital-71089914-uhmep",
  "clientAddress": "ip_address",
  "client_id": "nihdi-hospital-71089914-uhmep",
  "userProfile": {
    "organizations": [
      {
        "hospital": {
          "nihii": "71089914",
          "name": "HOSPITAL UHMEP SMALS M2M TEST"
        }
      ]
    }
  }
}

```

```
}  
}
```

Example hospital token without pseudonymization scopes:

```
{  
  "exp": 1787928293,  
  "iat": 1787927993,  
  "jti": "trrtcc:5bf6b856-0924-b9ed-e1a8-a4c4f81830c6",  
  "iss": "https://api-acpt.ehealth.fgov.be/auth/realms/M2M",  
  "aud": [  
    "nihdi-hospital-71089914-uhmep",  
    "nihdi-uhmep-api",  
    "nihdi-pss-api"  
  ],  
  "sub": "36ee742e-68bd-4029-b3c6-a9323e43d63c",  
  "typ": "Bearer",  
  "azp": "nihdi-hospital-71089914-uhmep",  
  "resource_access": {  
    "nihdi-uhmep-api": {  
      "roles": [  
        "organization"  
      ]  
    },  
    "nihdi-pss-api": {  
      "roles": [  
        "pss"  
      ]  
    }  
  },  
  "scope": "openid iam:authz nihdi:pss nihdi:uhmep:organization profile",  
  "clientHost": "ip_address",  
  "preferred_username": "nihdi-hospital-71089914-uhmep",  
  "clientAddress": "ip_address",  
  "client_id": "nihdi-hospital-71089914-uhmep",  
  "userProfile": {  
    "organizations": [  
      {  
        "hospital": {  
          "nihii": "71089914",  
          "name": "HOSPITAL UHMEP SMALS M2M TEST"  
        }  
      }  
    ]  
  }  
}
```

2.3.5. IAM.Exchange Security Commitment

The signature of these documents is required:

- [IAM eXchange Annex A Security commitment from the Trusted Platform](#)
- [IAM Connect Token Exchange Security Commitment in the context of pseudonymization](#)

2.3.6. Application Identifier

Your application identifier must be sent with each request via the User-Agent header. The value must conform to the following regular expression:

```
[a-zA-Z0-9-]*+\[ [a-zA-Z0-9-]*+\/[0-9a-zA-Z-_.]*+ \] (?:-[a-zA-Z0-9-]+)*+
```

Example value:

- YourCompanyName[YourSoftwarePackage/1.2.3]-YourConnector

2.3.7. AttributeAuthority

Recip-e integrates with AttributeAuthority to verify the patient's informed consent, therapeutic exclusions and therapeutic relations according to the regulations. More information can be found here: [NL](#), [FR](#).

Patients can use the MyHealth webapp to:

- Activate their informed consent
 - [ACC](#)
 - [PROD](#)
- Manage their therapeutic relations
 - [ACC](#)
 - [PROD](#)
- Manage their therapeutic exclusions
 - [ACC](#)
 - [PROD](#)

2.3.8. Break The Glass

Pharmacists can use Break The Glass on MedicationRequest search to bypass the Therapeutic Relation and/or Exclusion check. Technically, it's done by adding these search parameters:

```
bypassTherapeuticLinkApproved=true
bypassTherapeuticExclusionApproved=true
bypassTherapeuticLinkReason=<justification>
```

Use bypassTherapeuticLinkApproved=true to approve bypassing the absence of a therapeutic relationship. Use bypassTherapeuticExclusionApproved=true to approve bypassing a therapeutic exclusion. If either flag is true, a reason is mandatory.

Example:

```
POST /MedicationRequest/_search
Content-Type: application/x-www-form-urlencoded

patient=urn:be:fgov:ehealth:pseudo:v1:...&
status=active&
bypassTherapeuticLinkApproved=true&
bypassTherapeuticExclusionApproved=true&
bypassTherapeuticLinkReason=Emergency situation
```

3. FHIR Interactions

As mentioned before we follow the FHIR standards. It provides default resource definitions, created by HL7. In Belgium, eHealth (together with all parties stated above) defines specific FHIR profiles according to the Belgian requirements. The implementation guide for these profiles can be found here: <https://www.ehealth.fgov.be/standards/fhir/index.html>.

Recip-e supports only FHIR R4 as this is also the version used in the implementation guide.

3.1. CapabilityStatement

The server's capabilities of Recip-e will always be exposed via a FHIR CapabilityStatement. This contains both the supported operations as the supported profiles of the implementation guides. The optional and required search parameters for each resource are also mentioned in here (authorization required):

- [ACC](#)
- [PROD](#)

3.2. OpenAPI Definition

The OpenAPI specification of UHMEP can be found here:

- [ACC](#)
- [PROD](#)

3.3. Supported profiles

Recip-e supported the following FHIR profiles:

- [BeMedicationPrescription \(version 1.1.1\)](#)
 - Option to bundle it with [Consent](#) to manage prescription visibility
- [Task](#)
- [Bundle](#)

Note: The UHMEP platform supports more profiles than this. These are just the ones that are relevant for the Recip-e implementation.

3.4. Use of pseudonyms

Pseudonymization of the patient identifier in Recip-e codebase is carried as a FHIR Identifier on MedicationRequest.subject.

For integrators creating a MedicationRequest, the patient must be represented like this:

```
"subject": {
  "identifier": {
    "system":
"https://www.ehealth.fgov.be/standards/fhir/core/NamingSystem/ssin",
    "value": "urn:be:fgov:ehealth:pseudo:v1:some-value",
    "_value": {
      "extension": [{
        "url":
"https://www.ehealth.fgov.be/standards/fhir/infsec/StructureDefinition/be-ext-
pseudonymization",
        "extension": [
          { "url": "marker", "valueBoolean": true },
          { "url": "version", "valueInteger": 1 },
          { "url": "format", "valueCode": "direct" }
        ]
      }]
    }
  }
}
```

Rules:

- marker must always be true
- The version communicated as “version” in the extension, must match the one communicated in the “value” prefix. version = 1 corresponds to the value prefix “urn:be:fgov:ehealth:pseudo:v1:”, meaning
- Version 1 or prefix “urn:be:fgov:ehealth:pseudo:v1:” means in-transit data is sent BASE64 encoded.
- Version 2 or prefix “urn:be:fgov:ehealth:pseudo:v2:” means in-transit data is sent SEC1 encoded.
- If the version or value prefix are missing, the Recip-e server will try to determine based on the value which decryption algorithm to use. This is just fallback behaviour, so integrators are encouraged to always send the exact version of the algorithm to be used.

3.4.1. Pseudonymization Negotiation Headers

Use these headers to negotiate the encryption algorithm used to transmit in-transit pseudonyms.

3.4.1.1. Accept-Be-Pseudo

Clients may send the `Accept-Be-Pseudo` request header on Recip-e API calls when they want the response to contain patient pseudonyms in a specific supported in-transit format.

Supported values:

- `Accept-Be-Pseudo: urn:be:fgov:health:pseudo:v1` (default)
- `Accept-Be-Pseudo: urn:be:fgov:health:pseudo:v2`

3.4.1.2. Content-Be-Pseudo

Clients will receive the `Content-Be-Pseudo` response header telling the caller which pseudonymization version was used in the returned response.

3.4.2. Additional Documentation

Please find the FHIR documentation on pseudonymization here:

- [Pseudonymization Extension](#)
- [Content Negotiation for pseudonymization in FHIR](#)

3.5. Error codes

In case of issues, Recip-e will generate certain error codes and messages. Messages will be self-explanatory. Any informational, warning or error messages will be returned as `OperationOutcome`. In case of errors, the `OperationOutcome` will be the only response.

The error codes will be returned as part of the `urn:be:uhmep:mp:errors` system.

A list of all error codes can be found here:

- <https://github.com/smals-belgium/shared-digital-medication-prescription>

Example:

```
{
  "resourceType": "OperationOutcome",
  "issue": [
    {
      "severity": "error",
      "code": "business-rule",
      "details": {
        "coding": [
          {
            "system": "urn:be:uhmep:mp:errors",
            "code": "PRESCRIPTION_NOT_FOUND"
          }
        ]
      },
      "diagnostics": "Prescription not found."
    }
  ]
}
```

Below is some guidance to interpret the error codes and messages correctly.

First of all Recip-e follows the FHIR and HTTP specs as best as possible:

- 2xx codes are positive statuses
- 4xx statuses mean something is wrong with the request
- 5xx statuses mean something is wrong on the server

3.5.1. External services

Recip-e is dependent on external services. If one of these external services is down or a request to these services fails, a 500, 503 or 504 error will be thrown. It is very important to read the response body to derive if the problem is related to Recip-e itself or to an external service.

3.5.2. Validations

There are a lot of validations done on the request before it is accepted by the server. By failure of validations the server will respond with a 4xx error code with an error message that will say what is wrong with the request:

- 400 Bad Request: when there is a semantic problem with the request itself
- 422 Unprocessable Entity: when the requested resource is not in a valid state to make this request
- 404 Not Found: when the requested resource does not exist

3.5.3. Authentication and authorization

Everything that has to do with authentication or authorization will throw either a 401 or 403 error. To successfully register or consult a resource in Recip-e, the user will be verified by access control rules. Errors like software-white list (access based on client-id), role-based access matrix (access based on role) are self-evident. Recip-e also verifies therapeutic link, therapeutic exclusion and patient consent depending on the scenario. In case any of these checks fail an error will be thrown. Verify if the user applies to the necessary expected relationships. In case you think access should be granted, contact Recip-e support. If an error message is unclear, support can be contacted. It's important to add the full request and response in the communication. The response contains a Correlation-Id header which is the only way our support team can trace the request internally, so it should be mentioned in a support request.

3.5.4. Internationalization

For each error code, Recip-e also returns a default translated error message. These can be used by integrators that do not wish to provide their own translations. The language in which these are returned is steered by the `Accept-Language` request header, which can have the values:

- nl
- fr
- en (default)

4. Headers Summary

4.1. Request Headers

Header	Values	Required
Authorization	Bearer <access token>	Yes
User-Agent	<client application identifier>	Yes
Accept-Language	- nl - fr - en	No
Accept-Be-Pseudo	- urn:be:fgov:health:pseudo:v1 - urn:be:fgov:health:pseudo:v2	No
Content-Type	application/fhir+json	Yes
Accept	application/fhir+json	No
Correlation-Id	Any value supported by your tracing system. If absent, our system generates a UUID.	No

4.2. Response Headers

Header	Values	Required
Content-Be-Pseudo	- urn:be:fgov:health:pseudo:v1 - urn:be:fgov:health:pseudo:v2	Yes
Correlation-Id	<unique uuid for the request>	Yes
Content-Type	application/fhir+json	Yes

5. Business information

5.1. Business rules

Most of the business rules defined in the Implementation Guide but not all rules can be defined as constraints on FHIR elements.

Those rules are written down and explained below:

- When prescribing using an Actual Medicinal Product Package (AMPP) then the requested dispense quantity should be exactly 1 package.
If multiple packages are needed then multiple prescriptions have to be created.
- The expiration period of a prescription should be as short as possible and no longer than 1 year.
- The priority should always be set to "routine"
- The intent should always be set "order"
- Dates are used exactly as specified, and periods are inclusive.
This means that, for example, if a user searches for prescriptions with a specific date, only results with that exact date will be returned. When searching using a period, the results will have a date that is greater than or equal to the start date of the period and less than or equal to the end date of the period.
- It is possible to add optional CTI-extended coding to Medication using the system
"https://www.ehealth.fgov.be/standards/fhir/medication/NamingSystem/cti-extended-code"
An example of a Medication element with both CNK and CTI-extended is shown below:

```
{
  "resourceType": "Medication",
  "id": "med1",
  "code": {
    "coding": [
      {
        "system": "urn:be:samv2:type",
        "code": "amppackage"
      }, {
        "system": "https://www.ehealth.fgov.be/standards/fhir/medication/NamingSystem/cnk-
codes",
        "code": "1591163",
        "display": "2LEV globuli verp. eenmalig gebr. 30",
        "userSelected": true
      }, {
        "system": "https://www.ehealth.fgov.be/standards/fhir/medication/NamingSystem/cti-
extended-code",
        "code": "51737701",
        "display": "2LEV globuli verp. eenmalig gebr. 30",
        "userSelected": true
      }
    ]
  }
}
```

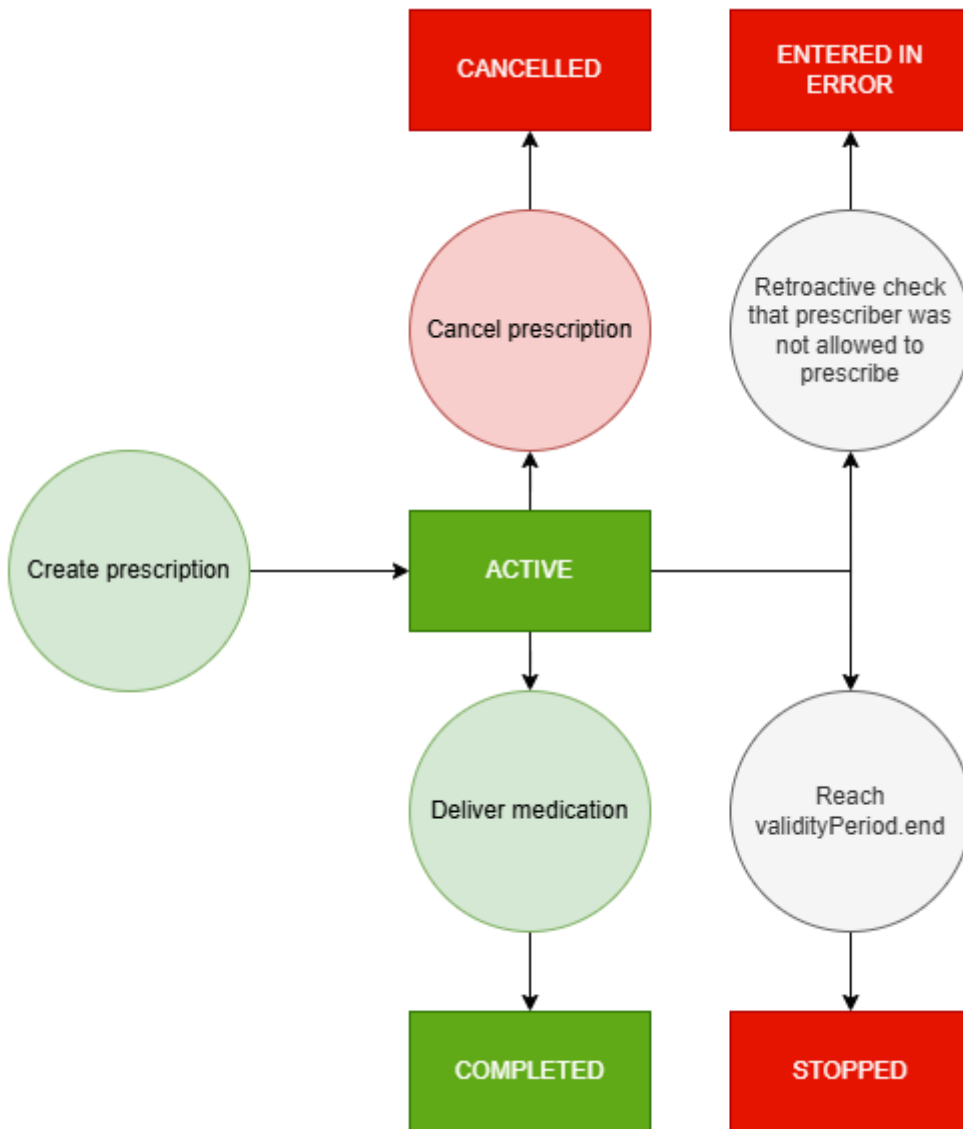
- It is required to provide the name, first name and address of the prescriber who created the prescription.
Optionally contactdetails such as phonenumber and emailaddress can also be provided.
An example of a Practitioner with cotnactdetails is shown below:

```

{
  "resourceType": "Practitioner",
  "id": "5d79c025-c727-427c-8fb1-7eb0cba34ac2",
  "meta": {
    "profile": [
      "https://www.ehealth.fgov.be/standards/fhir/core/StructureDefinition/be-practitioner"
    ]
  },
  "identifier": [
    {
      "system": "https://www.ehealth.fgov.be/standards/fhir/core/NamingSystem/nihdi",
      "value": "01234567890"
    }
  ],
  "name": [
    {
      "family": "Smals",
      "given": [
        "Rudy"
      ]
    }
  ],
  "telecom": [
    {
      "system": "phone",
      "value": "+32 16 555 30 00",
      "rank": 1
    },
    {
      "system": "pager",
      "value": "3000-101",
      "rank": 2
    },
    {
      "system": "email",
      "value": "rudy.smals@example.be",
      "rank": 3
    }
  ],
  "address": [
    {
      "line": [
        "Naamsestraat 101"
      ],
      "_line": [
        {
          "extension": [
            {
              "url": "http://hl7.org/fhir/StructureDefinition/iso21090-ADXP-streetName",
              "valueString": "Naamsestraat"
            },
            {
              "url": "http://hl7.org/fhir/StructureDefinition/iso21090-ADXP-houseNumber",
              "valueString": "101"
            }
          ]
        }
      ]
    }
  ],
  "city": "Leuven",
  "postalCode": "3000",
  "country": "BE"
}

```

5.2. Status flow



5.2.1. Explanation of the flow

- When a prescription is created then the status is set to ACTIVE.
- When an executor delivers the medication linked to the prescription then the status is set to COMPLETED.
- When a user cancels an ACTIVE prescription then the status is set to CANCELLED.
- Prescription rights can be revoke retroactively so at certain moments in the extended flow a check is done to ensure that the prescriber had the right to prescribe at the moment of prescription. When the prescribers right was revoked at the time of prescribing then the status is set to ENTERED IN ERROR.
- A validity end date is required. This date is automatically checked so that when that day is reached and passed the status of the prescription is automatically set to STOPPED.

5.3. Privacy flags

5.3.1. Explanation

Privacy flags determine, alongside informed consent, therapeutic relation, and therapeutic exclusion, the visibility of a prescription for prescribers and executors.

Two types of privacy flag exist:

1. The prescriber privacy flag, which limits the visibility of a prescription for prescribers
2. The pharmacy privacy flag, which limits the visibility of a prescription for executors

The prescriber privacy flag can be set to:

- All prescribers may see
- Only original prescriber
- Only original and GMR holder

An original prescriber will always be able to see a prescription. They cannot be blocked from viewing that prescription.

The pharmacy privacy flag can be set to:

- All pharmacies may see
- No pharmacy can see
- Only specific pharmacies may see

It is also possible to not set the privacy flags. When a privacy flag is not set, then it is considered to be open, so all related actors may see the prescription.

The privacy flags can be set by the prescriber when creating a prescription. They can set the prescriber privacy flags to any value they want. They can only set the pharmacy privacy flag to 'all pharmacies may see' or 'no pharmacy can see'.

The patient can also set the privacy flags. They have a full range of all the values of both privacy flags, and they can select a specific pharmacy to view their prescription.

Important to note is that when a patient sets the pharmacy privacy flag to 'only specific pharmacies may see' and selects a specific pharmacy, then the prescriber can see that the flag was set to 'only specific pharmacies may see,' but cannot have the info about which specific pharmacy is selected.

Circle Of Trust

A Circle-Of-Trust (COT) is a set of methodological and technical agreements between trusted parties that irrefutably secure information.

In this instance it's a set of agreements that is made between the hospital and eHealth. The COT determines everything about who can see which prescription so it decides itself about using privacy flags or not, the same with the informed consent, therapeutic relation, and therapeutic exclusion.

5.3.2. FHIR

Consent and Provision

The privacy flags are defined using the FHIR Consent element: [Consent - FHIR v4.0.1](#)

The Consent element has three fields which are required by FHIR and one which is required by our business rules. They are:

- **Status:** Indicates the state of the consent. Has to always be "active"
- **Scope:** Which of the four areas this Consent covers.
 - System: Has to always be "<http://terminology.hl7.org/CodeSystem/consentscope>"
 - Code: Has to always be "patient-privacy"
 - Display: Has to always be "Privacy Consent"
- **Category:** Classification of the Consent.
 - System: Has to always be "<http://terminology.hl7.org/CodeSystem/v3-ActCode>"
 - Code: Has to always be "INFA"
 - Display: Has to always be "Information access"
- **Policy:** Policies covered by this consent.
 - Authority: Has to always be "<https://www.riziv.fgov.be>"
 - Uri: Has to always be "<https://www.ehealth.fgov.be/ehealthplatform/file/cc73d96153bbd5448a56f19d925d05b1379c7f21/9d861ec73faaa1b588a5ed0587fa680f33c35a99/10-085-n376-recipe-en-webtoepassing-paris-gewijzigd-op-7-december-2021.pdf>"

Inside a Consent element, a Provision element is defined. The Provision element details which specific denies and permits are used for the prescription. When more than one deny or permit is needed, then a nested Provision element can be added to the main Provision element.

A provision element has three required fields:

- **Type:** How a rule statement is applied, such as adding additional consent or removing consent. Has to be “deny” or “permit”
- **Actor:** Who|what controlled by this rule (or group, by role). Pharmacy = “PHARMACY”, prescriber = “PROV”, GMD-holder = “GMD”
- **Data:** Data controlled by this rule. Contains a reference to the prescription using the RID

Prescriber privacy flag

Empty

It is possible to not set the prescriber privacy flag. When this is done, then it is considered that all prescribers may see the prescription.

All prescribers may see

To allow all prescribers to see the prescription, you need to set a provision with type “permit”, actor.role.code “PROV” and actor.reference a practionerRole with code “persphysician” without a practitioner element.

```
"provision": {
  "type": "permit",
  "actor": [ {
    "role": {
      "coding": [ {
        "system": "urn:be:uhmep:mp:CodeSystem:privacy-provision-role",
        "code": "PROV",
        "display": "PROV"
      } ]
    }
  } ],
  "reference": {
    "reference": "#pr1" // persphysician general
  }
},
"data": [ {
  "meaning": "instance",
  "reference": {
    "identifier": {
      "system": "http://ehealth.fgov.be/standards/fhir/medication/NamingSystem/be-ns-prescription",
      "value": "RID"
    }
  }
} ]
}
```

Only original prescriber

To allow only the original prescriber to see the prescription, you need to set a provision with type “deny”, actor.role.code “PROV” and actor.reference a practionerRole with code “persphysician” without a practitioner element.

```

"provision": {
  "type": "deny",
  "actor": [ {
    "role": {
      "coding": [ {
        "system": "urn:be:uhmep:mp:CodeSystem:privacy-provision-role",
        "code": "PROV",
        "display": "PROV"
      } ]
    },
    "reference": {
      "reference": "#pr1" // persphysician general
    }
  } ],
  "data": [ {
    "meaning": "instance",
    "reference": {
      "identifier": {
        "system": "http://ehealth.fgov.be/standards/fhir/medication/NamingSystem/be-ns-prescription",
        "value": "RID"
      }
    }
  } ]
}

```

Only original and GMD holder

To allow the original prescriber and the GMD holder to see this prescription, you need to set a provision with type “deny”, actor.role.code “PROV” and actor.reference a practionerRole with code “persphysician” without a practitioner element, and a provision with type “permit”, actor.role.code “AFFL” and actor.reference a practionerRole with code “persphysician” without a practitioner element.

```

"provision": {
  "type": "deny",
  "actor": [ {
    "role": {
      "coding": [ {
        "system": "urn:be:uhmep:mp:CodeSystem:privacy-provision-role",
        "code": "PROV",
        "display": "PROV"
      } ]
    }
  } ],
  "reference": {
    "reference": "#pr1" // persphysician general
  }
},
"data": [ {
  "meaning": "instance",
  "reference": {
    "identifier": {
      "system": "http://ehealth.fgov.be/standards/fhir/medication/NamingSystem/be-ns-prescription",
      "value": "RID"
    }
  }
} ],
"provision": [ {
  "type": "permit",
  "actor": [ {
    "role": {
      "coding": [ {
        "system": "urn:be:uhmep:mp:CodeSystem:privacy-provision-role",
        "code": "GMD",
        "display": "Global Medical Record Holder"
      } ]
    }
  } ],
  "reference": {
    "reference": "#pr2" // persphysician general
  }
} ],
"data": [ {
  "meaning": "instance",
  "reference": {
    "identifier": {
      "system": "http://ehealth.fgov.be/standards/fhir/medication/NamingSystem/be-ns-prescription",
      "value": "RID"
    }
  }
} ],
} ]
}

```

Pharmacy privacy flag

Empty

It is possible to not set the pharmacy privacy flag. When this is done, then it is considered that all pharmacies may see the prescription.

All pharmacies may see

To allow all pharmacies to see the prescription, you need to set a provision with type “permit”, actor.role.code “PHARMACY” and actor.reference a practitionerRole with code “orgpharmacy” without a practitioner element.

```
"provision": {
  "type": "permit",
  "actor": [ {
    "role": {
      "coding": [ {
        "system": "urn:be:uhmep:mp:CodeSystem:privacy-provision-role",
        "code": "PHARMACY",
        "display": "Pharmacy"
      } ]
    }
  } ],
  "reference": {
    "reference": "#pr1" // orgpharmacy general
  }
},
"data": [ {
  "meaning": "instance",
  "reference": {
    "identifier": {
      "system": "http://ehealth.fgov.be/standards/fhir/medication/NamingSystem/be-ns-prescription",
      "value": "RID"
    }
  }
} ]
}
```

No pharmacy can see

To deny all pharmacies from seeing the prescription, you need to submit a provision with the type “deny”, actor.role.code “PHARMACY” and actor.reference a practitionerRole with code “persphysician” without a practitioner element.

```
"provision": {
  "type": "deny",
  "actor": [ {
    "role": {
      "coding": [ {
        "system": "urn:be:uhmep:mp:CodeSystem:privacy-provision-role",
        "code": "PHARMACY",
        "display": "Pharmacy"
      } ]
    }
  } ],
  "reference": {
    "reference": "#pr1" // orgpharmacy general
  }
},
"data": [ {
  "meaning": "instance",
  "reference": {
    "identifier": {
      "system": "http://ehealth.fgov.be/standards/fhir/medication/NamingSystem/be-ns-prescription",
      "value": "RID"
    }
  }
} ]
}
```

Only specific pharmacies can see

To allow only a specific pharmacy to see the prescription, you set a provision with type "deny", actor.role.code "PHARMACY" and actor.reference a practitionerRole with code "orgpharmacy" without a practitioner element, and a provision with "deny", actor.role.code "PHARMACY", and actor.reference a practitioner role with code "orgpharmacy", with a practitioner element where a pharmacist is identified by their NIHI.

```
"provision": {
  "type": "deny",
  "actor": [ {
    "role": {
      "coding": [ {
        "system": "urn:be:uhmep:mp:CodeSystem:privacy-provision-role",
        "code": "PHARMACY",
        "display": "Pharmacy"
      } ]
    }
  } ],
  "reference": {
    "reference": "#pr2" // orgpharmacy general
  }
},
"data": [ {
  "meaning": "instance",
  "reference": {
    "identifier": {
      "system": "http://ehealth.fgov.be/standards/fhir/medication/NamingSystem/be-ns-prescription",
      "value": "RID "
    }
  }
} ],
"provision": [ {
  "type": "permit",
  "actor": [ {
    "role": {
      "coding": [ {
        "system": "urn:be:uhmep:mp:CodeSystem:privacy-provision-role",
        "code": "PHARMACY ",
        "display": "Pharmacy"
      } ]
    }
  } ],
  "reference": {
    "reference": "#pr3" // orgpharmacy with NIHI
  }
} ],
"data": [ {
  "meaning": "instance",
  "reference": {
    "identifier": {
      "system": "http://ehealth.fgov.be/standards/fhir/medication/NamingSystem/be-ns-prescription",
      "value": "RID"
    }
  }
} ],
} ]
}
```